

PlaceDAR User Manual

1. INTRODUCTION

This document serves as the official user manual for the PlaceDAR software, developed in response to the call for tender for the public procurement of a new MSSR radar implementation at Nice Côte d’Azur Airport.

PlaceDAR is an RF analysis and simulation tool designed for air navigation surveillance systems. This includes Primary Surveillance Radar (PSR), Monopulse Secondary Surveillance Radar (MSSR), and ADS-B sensors. The software enables the identification and visualization of all valid geographical locations for MSSR radar installation. This can be done through customizable parameters within the interface.

The tool provides optical and RF coverage analysis for selected locations, supporting radar siting decisions.

This software is intended for use by the contracting authority and authorized personnel involved in radar system design and evaluation.

2. SYSTEM REQUIREMENTS

Supported Operating Systems

PlaceDAR is supported on the following operating systems:

- Windows
- MacOS: PlaceDAR is supported, but users are advised not to open the 3D terrain map, as this feature can lead to crashes caused by excessive CPU load.

Software Requirements

The following software components are required:

- Python: Version 3.9 or higher recommended
- pip: Latest version recommended

3. INSTALLATION AND SETUP

Software Installation

Step 1: Install Python (if not already done)

1. Download Python from the official Python distribution.
2. Install Python version 3.9 or later.
3. Ensure the option “Add Python to PATH” is selected during installation.

Verify installation:

```
python --version
```

Step 2: Install Dependencies

From the PlaceDAR installation directory:

```
pip install -r requirements.txt
```

Installation time may vary depending on system performance and network speed.

Step 3: Launch the Application

Run the file main.py file

After a few seconds, the PlaceDAR graphical user interface will be ready to use.

4. USING PlaceDAR

The PlaceDAR graphical user interface provides an interactive environment for the visualization, analysis, and evaluation of radar positioning scenarios. The main display supports both 2D and 3D visualization modes and allows users to interact with geographic data, define candidate radar locations, and assess line-of-sight and coverage constraints.

The visual is the map of the Nice area with longitude on the x-axis, latitude on the y-axis, and elevation colourmap. To further the scope of the application, the user has the option to switch to a map of Bhutan. While this map shows geographical terrain and allows line-of-sight calculations, it does not have any implemented constraints.

4.1 Display Modes and Navigation

The 2D mode provides a plan-view representation of the area of interest. Users can navigate the 2D map by:

- Zoom in using the lens icon in the tool bar
- Zoom out using the home icon in the tool bar
- Pan by using the cross icon in the tool bar

The 3D mode provides a three-dimensional visualization of the terrain and obstacles. The altitude scale lets the user vertically stretch the map to accentuate the terrain elevation. Users can navigate the 3D map by:

- Zoom in by placing the mouse on the target point and pressing the key f
- Fully zoom out by pressing the key r
- Pan view by pressing shift and left click
- Tilt view by pressing control and left tilt

Users can switch between 2D and 3D modes using the display mode selector in the toolbar.

4.2 Airport Visualization

The airport is plotted automatically on the map and labeled. This reference point remains visible in both 2D and 3D mode and serves as a spatial reference for all positioning analyses.

4.3 Map Layers Control

Various layer selection drop-down menus allow users to control which data layers are displayed on the map. Layers can be enabled or disabled independently.

The constraints layers are calculated using the parameters specified by the user through the 'Edit constraints' buttons. Note that the distances are measured in meters.

Constraints layers:

- Road boundaries: shades in red all the map areas that are not within the maximum specified distance from a road
- Building boundaries: shades in red all the map areas that are closer to a dwelling than the minimum specified distance
- Electric points boundaries: shades in red all the map areas that are not within the maximum specified distance from an electrical access point
- Country boundaries: shades in red all the map areas that are not French territory
- Sea boundary: shades in red all the map areas that represent the sea
- Maximum distance to airport boundary: shades in red all the map areas that are not within the maximum specified distance from Nice Airport
- Nice boundary: shades in red all the map areas that represent the commune of Nice

Elements layers:

- Constraints valid points: displays on the map all the points that satisfy all the constraints listed above in the constraints layers
- Constraints airport LOS valid points: displays on the map all the constraints valid points that also have a direct line of sight to the Nice Airport
- Electrical points: displays all the electrical access points
- Roads: displays all the roads that are accessible to heavy duty vehicles
- Buildings: displays all the dwellings

4.4 Radar Point Management

Adding Points

Radar candidate points can be added using two methods:

- Interactive selection: select 'Add points on map' on the toolbar and then click on the map to place the point.
- Coordinate entry: enter geographic coordinates (latitude, longitude, altitude) via the 'Add point using coordinates' button on the right panel. Note that the coordinate system used is WGS-84.

Each point is displayed on the map and added to the saved points list.

Right-clicking on any point on the saved points list allows that point to be renamed, deleted, or calculate its flight-level line-of-sight.

Saving and Loading Points

- The saved points can be saved to a file for later reuse using the 'Save points to file' button on the right panel. A single point can be saved to a file by right-clicking on any point on the saved points list.
- Previously saved point files can be loaded into a project using the 'Load points from file' button on the right panel.

Displaying Candidate Radar Positions

This can be done either through the 'Get potential radar position points (2 points)' on the right panel, or by selecting 'Constraints airport LOS valid points' from the elements layers in the toolbar.

4.5 Flight Level Line-of-Sight (FL LOS)

Loading FL LOS Data

Users may load existing FL LOS datasets via the 'Load flight level LOS from file' button on the right panel. Loaded FL LOS data are displayed as dedicated layers on the 2D & 3D maps.

Calculating FL LOS

For a point, PlaceDAR can calculate the FL LOS by right-clicking on any point on the saved points list. Note that a flight level corresponds to an altitude (in feet) of 100 times the value of the flight level. The user must provide the desired flight levels (e.g. 10, 50,100) and the

line-of-sight parameters (ray casting step size, radar antenna height, number of azimuthal angles, and LOS cutoff binary search distance delta). The resulting FL LOS and coverage maps are visualized directly in the main display. The right panel then displays the coverage percentage for each flight level and allows the user to select which flight level they which to visualize on the map.

4.6 Export to KMZ and Visualization in Google Earth

Exporting to KMZ

Once FL LOS data has been loaded or calculated, the analysis results, including radar candidate points and FL LOS data can be exported to a KMZ file using the 'export as kml/kmz' button in the toolbar.

Viewing in Google Earth

To visualize the exported data in Google Earth:

1. Open Google Earth.
2. Select New and then New map
3. Select File → Open.
4. Load the exported KMZ file.
5. Enable or disable layers as required.

6. USER ERRORS

Invalid Input

If a user-entered parameter value is not accepted or does not trigger an update of the map, the value is outside the allowed range. The user must enter a valid value within the specified limits.

Module not found or does not exist

One of the required modules has not been correctly installed. Please ensure that all the modules in requirements.txt have been correctly installed.